Todd Takala

Lead Data Scientist, Machine Learning Engineer, AI/ML of IoT and Physical Systems
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Summary

Lead Data Scientist and MLOps expert with a mechanical engineering and reliability background, specializing in turning complex sensor and operational data into production-grade machine learning solutions that improve physical asset reliability. Focused on preventing failures, maximizing uptime, and driving cost savings by optimizing predictive analytics and asset management.

Professional Experience

Caterpillar Inc.
Lead Data Scientist — Peoria, IL 09/2022 — Present

- Apply deep service and engineering expertise, plus data science, machine learning, and AI, to improve quote accuracy, optimize lifecycle cost models, and enhance labor forecasting for heavy equipment repairs, drawing on experience as a client-facing engineering manager, design engineer, and mechanic.
- Designs and implements resilient data solutions using Python, Snowflake, SQL, R, Power BI, Tableau, SnapLogic, Docker, and AWS, including an AI platform that forecasts heavy machinery repair times. This platform increased warranty labor coverage by 375%, received a CEO Award for strategic impact, and leverages machine learning, rigorous testing, and GitHub CI/CD to boost customer satisfaction and promote fair, dependable warranty claim outcomes.
- Led development and deployment of advanced ML models (using Docker, PyTorch, TensorFlow, etc.) that improved repair-time prediction accuracy by 40%, saving $4M by removing the need for physical time studies.
- Collaborate closely with senior leaders and stakeholders to simplify technical ideas into business benefits and promote scalable agile practices. This method has led to the successful adoption of projects and alignment with strategic goals for maximum enterprise impact.

Amazon.com
Reliability Analytics Manager — Tempe, AZ 02/2022 — 09/2022

- Applied advanced analytics, statistics, and mathematical modeling to enhance the efficiency and reliability of autonomous warehouse robots, which significantly improved operational oversight and decision-making.
- Created groundbreaking KPIs that offer detailed insights into robot efficiency and authored a white paper on using Weibull analysis for reliability design.
- Led a team of BI engineers and data analysts to develop a Shiny R pipeline on AWS for real-time global monitoring of robot status at Amazon fulfillment centers. This project improved operational visibility, sped up incident responses, and boosted data analytics for proactive maintenance and reliability.
- Engineered robust data management solutions using AWS Redshift and PostgreSQL to support predictive models and large-scale data analysis efforts, driving improved reliability predictions.

Empire-Cat
Data Engineering Manager — Mesa, AZ 05/2021 — 09/2022

- Develop AI models for time-series forecasting using Python, scikit-learn, TensorFlow, keras, R, tidyverse, ggplot2, and SQL to enhance demand forecasting and minimize stock-out risks in the mining business through MLOps integration.
- Oversaw a strategic revamp of the component exchange program, creating a scalable, data-driven infrastructure to improve efficiency and data analysis. Implemented data warehousing and ETL/ELT pipelines to enhance supply chain visibility and reliability, supporting decision-making with accurate insights.
- Efforts focused on creating best practices and governance frameworks in data engineering, emphasizing data quality and metadata management, to ensure reliable data foundations for analytics and reporting.

Empire-Cat
Reliability Engineering Manager — Mesa, AZ 01/2017 — 04/2021

- Achieves operational excellence through innovative predictive analytics and robust engineering solutions, aimed at enhancing the reliability, durability, and maintenance strategies of mining equipment.
- Developed regression machine learning models using R to forecast engine reliability and maintenance needs, which optimized maintenance processes and resulted in savings exceeding $62 million within the first 90 days.
- Led a team to foster collaboration and deliver expert technical support for Empire-Cat's mining clients, ensuring effective issue resolution and high customer satisfaction.
- Led the RCFA program using a data-driven method to identify and solve systemic issues, strengthening Empire FMEA and 6 Sigma processes. This improved product reliability, reduced field failures, and enhanced quality and safety, earning Caterpillar's recognition for industry best standards adherence.
- Enhanced engineering decisions and maintenance strategies through advanced data techniques with R, tidyverse, Python, pandas, scikit-learn, and SQL to ensure reliable and high-performing mining equipment.

Empire-Cat
Lead Reliability Engineer — Mesa, AZ 06/2016 — 12/2016

- Developed a reproducible research framework using R, SQL, and Weibull analysis to enhance product reliability. This approach, adopted by Caterpillar, streamlined processes and set a new industry standard.
- Develop and oversee a comprehensive failure analysis program that utilizes data-driven insights to enhance operational performance. Increased the physical availability of an aging product line from 80% to 90% through specific interventions, which greatly improved fleet operational efficiency and focused on remediating persistent major component issues.
- Enhanced the technical excellence of products and established Empire-Cat as an industry leader by promoting continuous improvement and setting standards for reliability engineering.

Empire-Cat
Technical Communications Manager — Mesa, AZ 12/2012 — 06/2016

- The text highlights the expertise in linking customer feedback with engineering innovation through leading a team of technical communicators. Leadership skills are used to synthesize client insights, identify key areas for product development, and drive performance improvements to enhance customer satisfaction.
- Integrating DMAIC Six Sigma with advanced computer science identifies and implements improvement opportunities, achieving $1.2 billion in savings and efficiencies over six years.
- A condition monitoring application was developed using R and SQL to automate fluid sample analysis for Empire-Cat, eliminating the need for costly software licenses and saving an estimated $100,000 annually. It provides crucial data for proactive maintenance and reliability assessments.
- Developed a Python and MS-SQL based tracking system that transformed fragmented data into actionable insights, speeding up evaluation and improving data-driven decisions in product development.
- Instrumental in advancing Empire-Cat's technical and operational excellence by integrating strategic leadership, creative problem-solving, and a customer-focused approach.

Empire-Cat
Reliability Engineer — Mesa, AZ 01/2012 — 12/2016

- Led the Product Problem Management program, which improved operational efficiency and optimized maintenance spending on equipment reliability in the mining fleet. This initiative systematically identifies, analyzes, and resolves recurring failures in mining haul trucks, significantly enhancing equipment reliability, reducing unplanned downtime, and boosting overall operational efficiency.
- Worked with cross-functional engineering teams to design and execute experiments using engineering and statistical methods, like Taguchi and response surface methodology. These efforts addressed critical failure modes and performance limitations, improving mining equipment's safety, durability, and performance.

Career Note

Additional tenure as a Technical Consultant & Technical Communicator at Road Machinery LLC; Research and Development Design Engineer at Komatsu America; Engineering Intern at X-tek Mining Services; Heavy Equipment Operator at Ryan Central Inc.; Heavy Equipment Mechanic at Roland Machinery; and Apprentice Heavy Equipment Mechanic at Harry W. Kuhn, Inc. details are available upon request.

Certification & Technical Registration

- **Arizona State Board of Technical Registration:** EIT, Mechanical Engineering, Registration No. 10670
- **Canonical:** Ubuntu Linux Professional Certificate
- **Data Camp:** AI Fundamentals Certificate • Certified SQL Associate, Data Scientist Professional Certificate
- **GitHub:** Career Essentials in GitHub Professional Certificate
- **IBM:** Advanced Deep Learning Specialist • RAG and Agentic AI Professional Certificate • Deep Learning with PyTorch, Keras and Tensorflow Professional Certificate
- **Komatsu North America:** Certified Technical Communicator
- **NCEES:** Passed FE Exam
- **UC, Berkeley:** The Science of Happiness at Work Professional Certificate
- **US Department of Labor:** Operating Engineer Vocational Certificate

Education

Arizona State University

Bachelor of Science in Mechanical Engineering Tempe, AZ

Excel at distilling complex physical systems into mathematical models and converting real-world engineering challenges into data-driven solutions. An engineering background combines deep technical expertise with rigorous problem-solving capabilities and attention to detail. This alignment supports advanced computational and analytical positions, particularly in Lead Data Scientist and Machine Learning Engineer roles.

Hard Skills

A/B Testing, Agentic AI, Agile Methodology, AI/ML, Anomaly Detection, Artificial Intelligence, AWS, BASH, CI/CD, Computer Science, RDBMS, Database Administrator, Data Science, DBT, Deep Learning, Design of Experiments, DevOps, Docker, Embedding, ETL, Fine Tuning, Advanced Excel, Financial Forecasting, Generative AI, Git, Linear Algebra, Large Language Models, LLM, Mathematics, Natural Language Processing, NLP, Pandas, Power BI, Public Cloud, Python, PyTorch, R, RAG, Research, Sagemaker, Scikit-learn, Sagemaker, Scripting Language, SDLC, Simulation, SnapLogic, Snowflake, SQL Queries, Statistical Analysis, Streamlit, SQL, Tableau, TensorFlow, Transact SQL, Transformers, Version Control

Soft Skills

Adaptability, Business Requirements, Coaching, Communication Skills, Continuous Improvement, Creative Problem Solving, Data Driven, Decision making, Growth Mindset, Interviewing, Leadership Experience, Life Long Learning, Listening, Mentoring, Methodical, Multicultural Collaboration, Open minded, Problem Solving, Professional Services, Presentations, Public Speaking, Strategic Thinking, Team Building, Thought Leadership, Willing to Learn